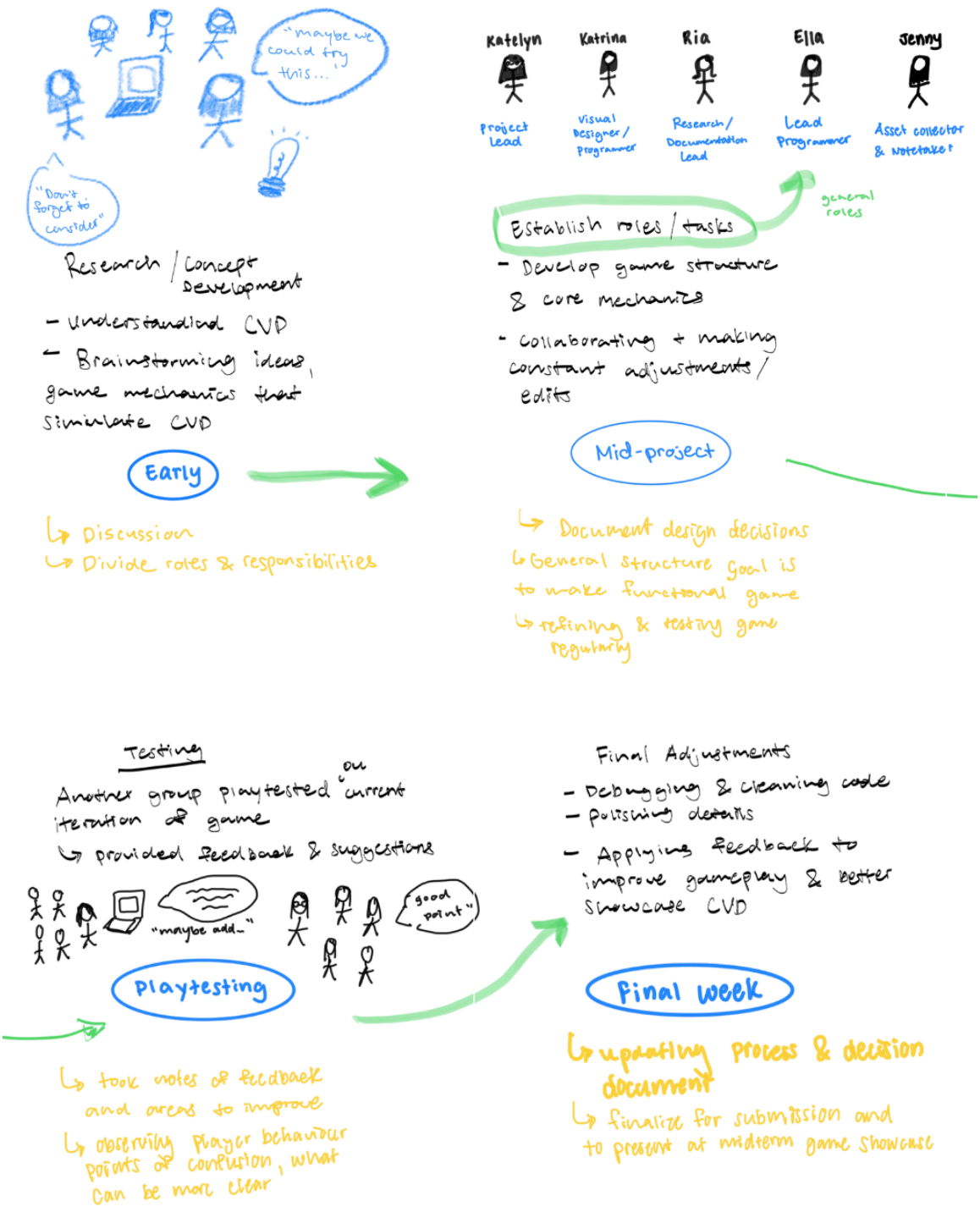
A2 Process & Decision Documentation

Process Overview Visualization



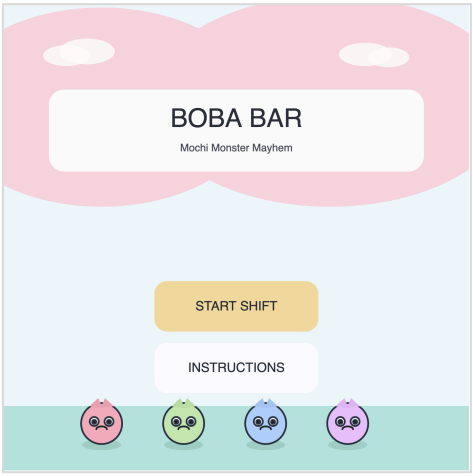
Project/Assignment Decisions

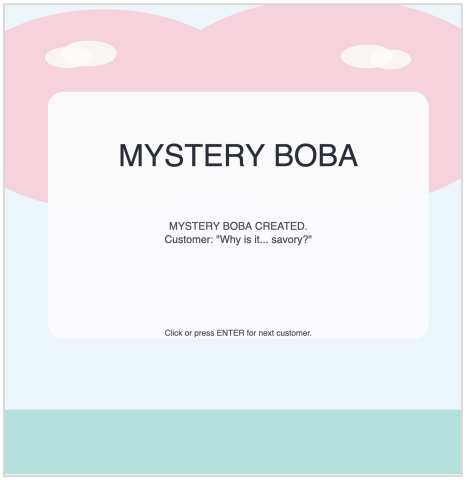
In Assignment 2, we developed our mid-term game prototype aimed at teaching non-colour blind individuals about colour vision deficiency and simulating what it would be like to possess it. GenAI was used to support the development of our game code, ensuring we maintained a consistent and smooth gameplay overall.

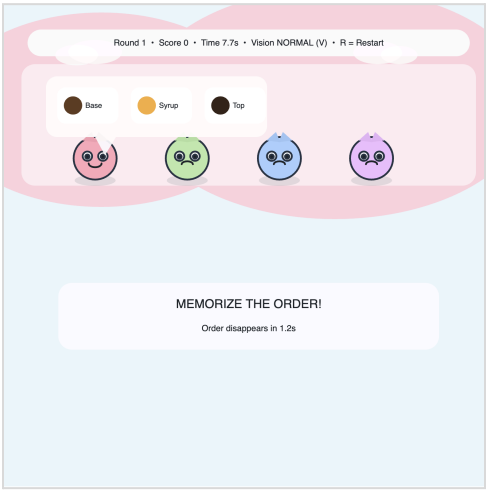
- Playtesting the game multiple times, noticed alignment issues with certain text and shapes such as the tea, syrup and topping section
    - Text was not properly aligned nor within box perimeters
- Group discussions allowed us to share ideas on how to make the game more visually appealing, engaging, and functional
    - Adjusting font, naming/text, colours (certain colours are used to emphasize importance, such as the “order disappears in…” text)

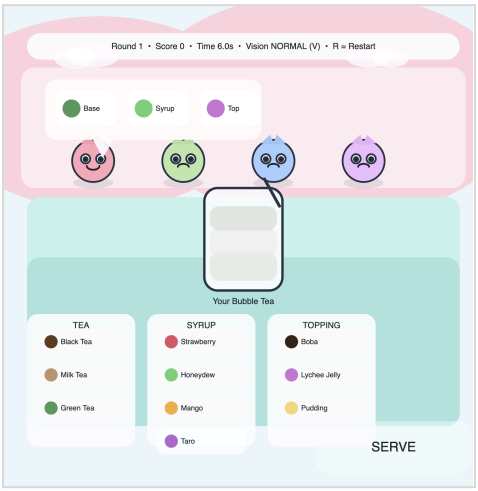
Role-Based Process Evidence

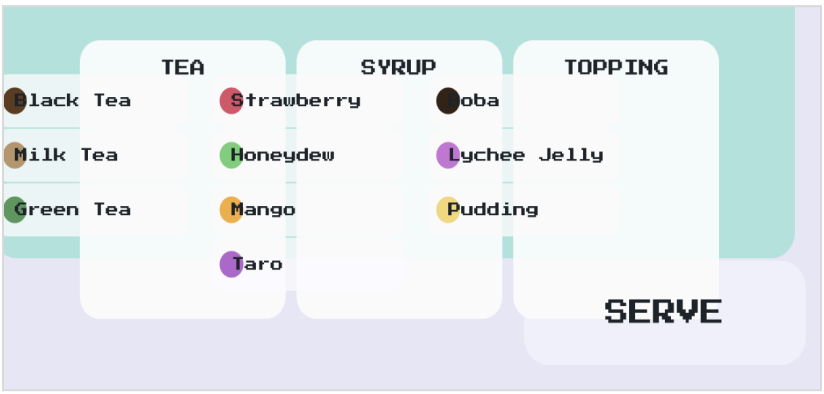
(Initial Game)



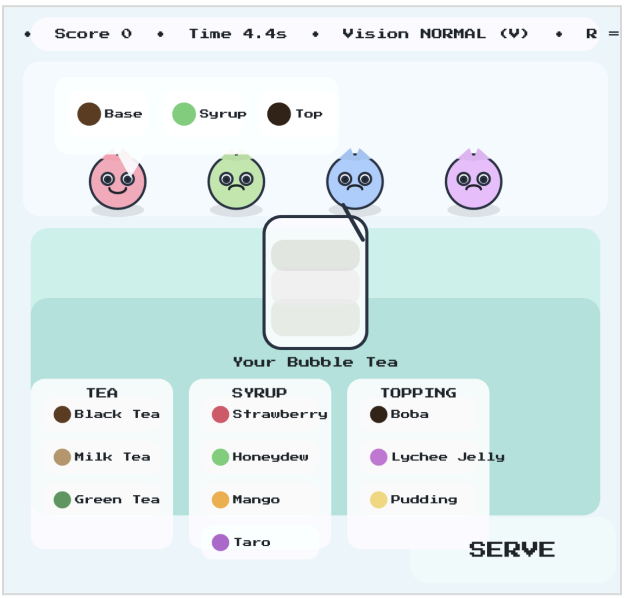


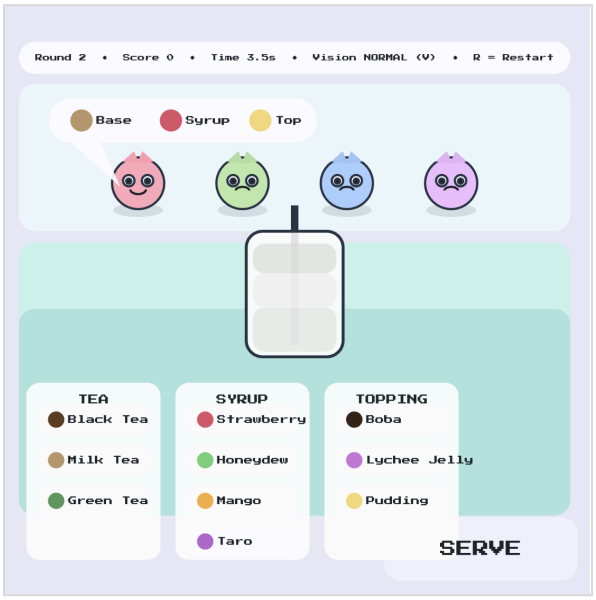


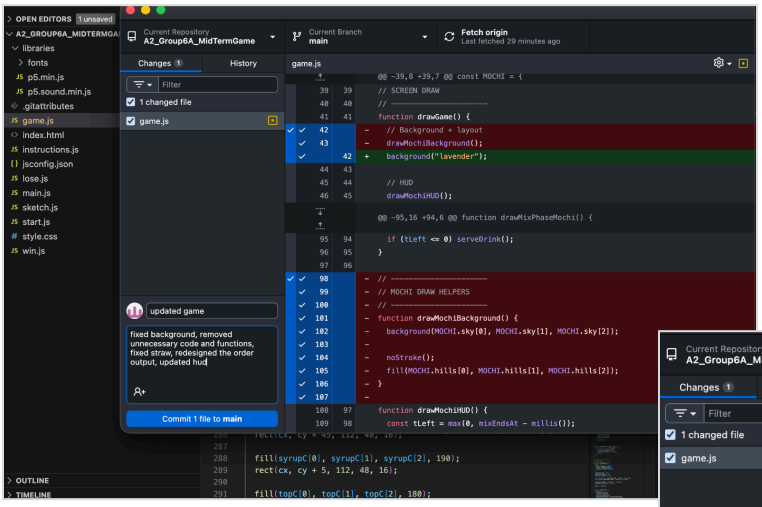


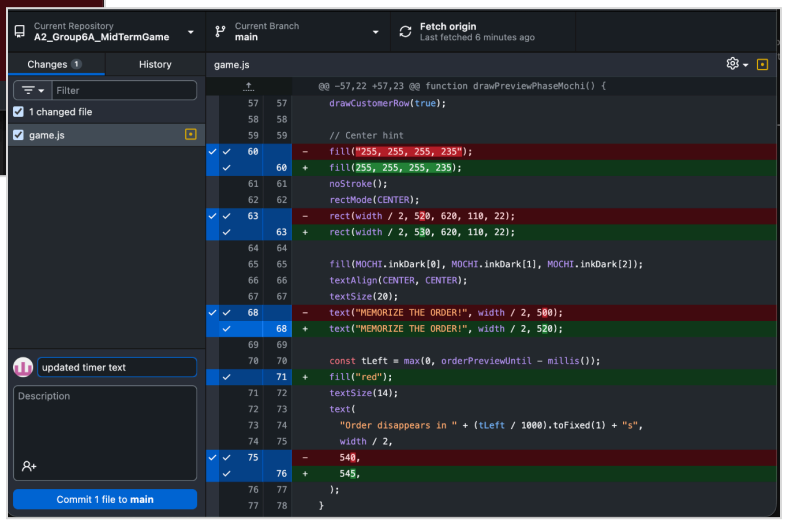


(Iteration/progress)









Goal of Work Session

- Brainstorm game concept and mechanics related to colour vision deficiency
- Build and code game prototype to reflect proposed ideas
- Debug and refine code to ensure proper level progression
- Adjust visual elements and overall appearance
- Revise key mechanics based on playtesting feedback
- Fix areas or errors affecting gameplay experience
- Document suggestions and decisions for future reference

Tools, Resources, or Inputs Used

- ChatGPT 5.2
- Lecture notes and class examples
- Teammates / peer support
- Playtesting feedback

GenAI Documentation

Date Used: Mar 4, 2026

Tool Disclosure: ChatGPT 5.2

Purpose of Use: To brainstorm ideas relevant to our disability and help develop code for game prototype.

Summary of Interaction: We provided the tool with multiple game concepts similar to Hue (2016 puzzle-platform game), then helped convert the chosen concept into a single level loop and we proposed a Mochi Monster-inspired UI layout using only p5 shapes.

Human Decision Point(s): We made the final visual decisions of the game by editing and cleaning shape and text alignment, font type, and colours for a more cohesive and appealing look.

Integrity & Verification Note: We checked GenAI for accuracy, bias, and made sure it was appropriate to fit for this assignment. We made sure the output aligned with the class materials and verified it by rereading the assignment instructions and comparing them to GenAI suggestions. Any content that was beyond what we learned in class was removed and replaced.

Scope of GenAI Use: GenAI was mainly used during the ideation/starting stage and for outlining the basic structure of the game. The final design choices, testing, and adjustments to the mechanics were done by us.

Limitations or Misfires: Limitations when clicking or interacting with certain actions. Sometimes the tool could not code certain actions properly or as we intended/expected, and this is where we had to debug.

Summary of Process (Human + Tool)

We explored several concepts that teach colourblindness through mechanics (puzzles, matching, reaction games) and chose bubble tea assembly because it supports memory, time pressure, and colour-based selection.

Decision Points & Trade-offs

We considered (1) a Hue-style platformer and (2) a bubble tea “order memory + mixing” game. Ultimately, deciding on a bubble tea inspired approach because it better supports short rounds, quick feedback, and a clear order UI like Mochi Monster. Additionally, through playtesting with our classmates, we received feedback in regards to the current UI and how the colour deficiency aspect could be more evident. Originally, we had two vision modes that allowed for the colours to toggle between “normal vision” mode and CVD (“colour vision deficiency”) mode. Through feedback, we were suggested to remove the modes and instead begin the game with “normal vision” which automatically switches to CVD once the order is displayed. In order for players to better notice the colour differences and experience a bit more challenge when distinguishing colours as a way to represent our disability.

Verification & Judgement

To verify our decisions and justify our changes, we all collectively playtested the game together. As each member took turns playing the game, we were able to find small bugs and visual aspects to be changed and enhance the gameplay. Notably, we found that some of the shapes and text were not aligning together and that the font was too basic, so we changed these factors and it made the overall game more smooth and visually appealing.

Limitations, Dead Ends, or Open Questions

Unfortunately, our UI did not work entirely as expected; the order would not disappear once the ingredient options/kiosk appeared. Impacting the gameplay since players were meant to memorize the order and colours before the kiosk is displayed, then attempt to select the correct ingredients. This error removes a level of difficulty and struggle, making it fairly simple to play and potentially less engaging. However, something we can surely improve on moving forward.

Appendix

Prompt:
For GBDA 302 Assignment 2 need a game that teaches non-colourblind players about colour blindness through mechanics, examples like Hue use colour mechanics group liked the Moshi Monster cooking mini-game style more idea: bubble tea shop game reason: colourful ingredients + mixing drinks fits gameplay also something our group likes in real life player sees drink order → has to recreate it mechanic idea: colours harder to distinguish (simulate colourblind confusion) need to keep it simple and within p5.js + class material.

GenAI prompt:
Some ways you could structure this idea:
Basic gameplay loop

- Show the player a bubble tea order for a short time (tea base + syrup + topping)
- Hide the order after a few seconds
- Player selects ingredients to recreate the drink
- Press a serve button to check if the drink is correct

Colourblind mechanic ideas

- Some ingredient colours become harder to tell apart
- Add a toggle that changes how colours appear
- Encourage players to rely on memory and position, not just colour

Simple UI layout

- Customer characters at the top showing orders
- Ingredient bins at the bottom (tea base, syrup, toppings)
- Cup in the middle where the drink is assembled
- Score or feedback after serving the drink

Technical approach (within p5.js basics)

- Use simple shapes (rect, ellipse) to draw ingredients
- Use variables to store the current order
- Use a timer with millis() for the preview phase
- Use mouse clicks to select ingredients